Status	Finished
Started	Friday, 30 October 2026, 2:21 PM
Completed	Friday, 30 October 2026, 2:28 PM
Duration	6 mins 55 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the output of the given below program?

```
t = (58, 47, 36, 25, 14, 3)
x = t[2:-1]
print(x)
```

a. (3,14,25)

b. (58,47,36,25)

c. (36, 25, 14) ✓

d. Error

Question **2**

Correct

Mark 1.00 out of 1.00

Choose the correct option.

a. In Python, a tuple can contain only strings as its elements.

b. In Python, a tuple can contain only integers as its elements.

c. In Python, a tuple can contain either string or integer but not both at a time.

d. In Python, a tuple can contain both integers and strings as its elements. ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Find the output of the given Python program?

```
t = (11, 3)
x = 3 * t
print(x)
```

a. (11,3)(11,3)(11,3)

b. [11,11,11,3,3,3]

c. (11,3,11,11,3,11,11,11,3)

d. (11, 3, 11, 3, 11, 3) ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Find the output of the given Python program?

```
t1 = (55, 44, 33, 22, 11)
x = [t1[i] for i in range(0, len(t1), 2)]
print(x)
```

a. [(55,33,11)]

b. (55,33,11)

c. ([55,33,11])

d. [55, 33, 11] ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What is the output of the following

```
set1 = {1, 2, 3, 4, 5}
set2 = {6, 7, 1, 3, 4, 8, 2, 5}

print(set1.issubset(set2))
print(set2.issuperset(set1))
```

a. True ✓

True

b. True

False

c. False

False

d. False

True

Question **6**

Correct

Mark 1.00 out of 1.00

What is printed when the following code is run?

```
tup = ('30', '3', '2', '8')
print(sorted(tup,reverse = True))
```

a. ['2', '3', '8', '30']

b. ['30', '8', '3', '2']

c. ['2', '3', '30', '8']

d. ['8', '30', '3', '2'] ✓

Question **7**

Correct

Mark 1.00 out of 1.00

What will be the output of following Python code?

```
set1={2,5,3}

set2={3,1}

set3={}

set3=set1&set2

print(set3)
```

a. {2,5,3,1}

b. {}

c. {3} ✓

d. {2,5,1}

Question **8**

Correct

Mark 1.00 out of 1.00

What is the output of the following code

```
aSet = {1, 'rec', ('cse', 'ece'), True}
print(aSet)
```

a. {'rec', 1, ('cse', 'ece'),True}

b. Error

c. {'rec', 1, ('cse', 'ece')} ✓

d. {'rec', True, ('cse', 'ece')}

Question **9**

Correct

Mark 1.00 out of 1.00

What is the output of the given below program?

my_t1 = (1, 2, 3, 4)

my_t1.append((5, 6, 7))

print(len(my_t1))

- a. 5
- b. 1
- c. Error ✓
- d. 2

Question **10**

Correct

Mark 1.00 out of 1.00

What is the output of the following union operation

```
set1 = {10, 20, 30, 40}
set2 = {50, 20, "10", 60}

set3 = set1.union(set2)
print(set3)
```

- a. {40, 10, '10', 50, 20, 60, 30} ✓
- b. {40, '10', 50, 20, 60, 30}
- c. SynatxError: Different types cannot be used with sets
- d. {40, 10, 50, 20, 60, 30}

Question **11**

Correct

Mark 1.00 out of 1.00

What is the output of the following code?

```
aTuple = (10, 20, 30, 40, 50, 60, 70, 80)
print(aTuple[2:5], aTuple[:4], aTuple[3:])
```

a. (30, 40, 50)(40, 50, 60, 70, 80)

b. (10, 20, 30, 40) (40, 50, 60, 70, 80)

c. (30, 40, 50) (10, 20, 30, 40) (40, 50, 60, 70, 80) ✓

d. (30, 40, 50) (10, 20, 30, 40)

Question **12**

Correct

Mark 1.00 out of 1.00

A python tuple can be created without using any parentheses. (True/False)

a. False

b. True ✓

Question **13**

Correct

Mark 1.00 out of 1.00

What will be the output of following Python code?

```
list1=[1,3,4,2]

x=list1.pop(2)

print(set([x]))
```

a. {2}

b. {1,3,2}

c. {1,3,4}

d. {4} ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What will the below Python code do?

```
set1={2,3}

set2={3,2}

set3={2,1}

if(set1==set2):

    print("yes")

else:

    print("no")

if(set1==set3):

    print("yes")

else:

    print("no")
```

a. No, No

b. Yes, Yes

c. Yes, No ✓

d. "==" is not supported for set in Python

Question **15**

Correct

Mark 1.00 out of 1.00

If a=(15,16,17,18,19,25), then a[1:-1] will be

Note :

a=(15,16,17,18,19,25)

print((a[1:-1]))

a. Error

b. (16,17,18)

c. (16,17,18,19) ✓

d. (25,19,18,17)

Status	Finished
Started	Friday, 30 October 2026, 2:30 PM
Completed	Friday, 30 October 2026, 2:39 PM
Duration	8 mins 32 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

There is a malfunctioning keyboard where some letter keys do not work. All other keys on the keyboard work properly.

Given a string text of words separated by a single space (no leading or trailing spaces) and a string brokenLetters of all distinct letter keys that are broken, return the number of words in text you can fully type using this keyboard.

Example 1:

Input: text = "hello world", brokenLetters = "ad"

Output:

1

Explanation: We cannot type "world" because the 'd' key is broken.

For example:

Input	Result
hello world ad	1
Faculty Upskilling in Python Programming ak	2

Answer: (penalty regime: 0 %)

```
text = input()
brokenLetters = input()

broken_set = set(brokenLetters.lower())
words = text.split(' ')

valid_words = 0
for word in words:
    if not any(char in broken_set for char in word.lower()):
        valid_words += 1

print(valid_words)
```

	Input	Expected	Got	
✓	hello world ad	1	1	✓
✓	Welcome to REC e	1	1	✓
✓	Faculty Upskilling in Python Programming ak	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Program to print all the distinct elements in an array. Distinct elements are nothing but the unique (non-duplicate) elements present in the given array.

Input Format:

First line take an Integer input from stdin which is array length n.

Second line take n Integers which is inputs of array.

Output Format:

Print the Distinct Elements in Array in single line which is space Separated

Example Input:

5

1 2 2 3 4

Output:

1 2 3 4

Example Input:

6

1 1 2 2 3 3

Output:

1 2 3

For example:

Input	Result
5 1 2 2 3 4	1 2 3 4

Answer: (penalty regime: 0 %)

```
n = int(input())

# Read n integers (handles both space-separated or new-line separated inputs)
elements = []
while len(elements) < n:
    elements.extend(map(int, input().split()))

# Preserve insertion order using dict.fromkeys()
distinct_elements = list(dict.fromkeys(elements))

print(*distinct_elements)
```

	Input	Expected	Got	
✓	5 1 2 2 3 4	1 2 3 4	1 2 3 4	✓
✓	6 1 1 2 2 3 3	1 2 3	1 2 3	✓
✓	5 11 22 11 22 11	11 22	11 22	✓
✓	10 1 2 3 4 5 1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

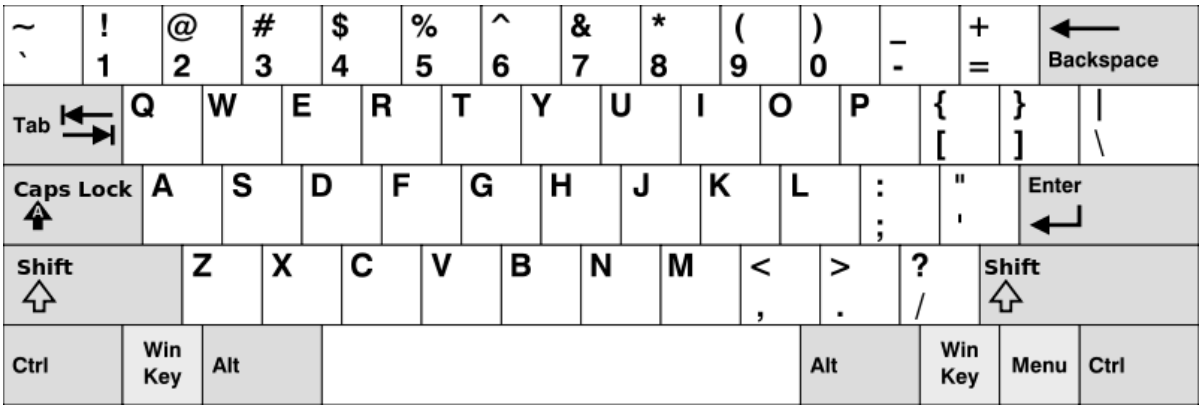
Correct

Mark 1.00 out of 1.00

Given an array of strings `words`, return *the words that can be typed using letters of the alphabet on only one row of American keyboard like the image below*.

In the **American keyboard**:

- the first row consists of the characters `"qwertyuiop"`,
- the second row consists of the characters `"asdfghjkl"`, and
- the third row consists of the characters `"zxcvbnm"`.



Example 1:

```
Input: words = ["Hello","Alaska","Dad","Peace"]
Output: ["Alaska","Dad"]
```

Example 2:

```
Input: words = ["omk"]
Output: []
```

Example 3:

```
Input: words = ["adsdf","sfd"]
Output: ["adsdf","sfd"]
```

For example:

Input	Result
4 Hello Alaska Dad Peace	Alaska Dad
2 adsfd afd	adsfd afd

Answer: (penalty regime: 0 %)

```
row1 = set("qwertyuiop")
row2 = set("asdfghjkl")
row3 = set("zxcvbnm")

n = int(input().strip())
```

```
6
7  result = []
8  for _ in range(n):
9      word = input().strip()
10     word_set = set(word.lower())
11     if word_set <= row1 or word_set <= row2 or word_set <= row3:
12         result.append(word)
13
14 if result:
15     for word in result:
16         print(word)
17 else:
18     print("No words")
```

	Input	Expected	Got	
✓	4 Hello Alaska Dad Peace	Alaska Dad	Alaska Dad	✓
✓	1 omk	No words	No words	✓
✓	2 adsfd afd	adsfd afd	adsfd afd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

You are given an integer tuple `nums` containing distinct numbers. Your task is to perform a sequence of operations on this tuple until it becomes empty. The operations are defined as follows:

1. If the first element of the tuple has the smallest value in the entire tuple, remove it.
2. Otherwise, move the first element to the end of the tuple.

You need to return an integer denoting the number of operations required to make the tuple empty.

Constraints

- The input tuple `nums` contains distinct integers.
- The operations must be performed using tuples and sets to maintain immutability and efficiency.
- Your function should accept the tuple `nums` as input and return the total number of operations as an integer.

Example:

Input: nums = (3, 4, -1)
Output: 5

Explanation:
Operation 1: [3, 4, -1] -> First element is not the smallest, move to the end -> [4, -1, 3]
Operation 2: [4, -1, 3] -> First element is not the smallest, move to the end -> [-1, 3, 4]
Operation 3: [-1, 3, 4] -> First element is the smallest, remove it -> [3, 4]
Operation 4: [3, 4] -> First element is the smallest, remove it -> [4]
Operation 5: [4] -> First element is the smallest, remove it -> []
Total operations: 5

For example:

Test	Result
`print(count_operations((3, 4, -1)))`	5

Answer: (penalty regime: 0 %)

Reset answer

```python
def count_operations(nums: tuple) -> int:
    ops = 0
    current_min = min(nums)

    while nums:
        ops += 1
        first = nums[0]

        if first == current_min:
            # Remove the first element using tuple slicing
            nums = nums[1:]
            if nums:
                # Recalculate minimum for the remaining tuple
                current_min = min(nums)
        else:
            # Move the first element to the end of the tuple
            nums = nums[1:] + (first,)

    return ops
```

	Test	Expected	Got	
✓	`print(count_operations((3, 4, -1)))`	5	5	✓
✓	`print(count_operations((1, 2, 3, 4, 5)))`	5	5	✓
✓	`print(count_operations((5, 4, 3, 2, 1)))`	15	15	✓
✓	`print(count_operations((42, )))`	1	1	✓
✓	`print(count_operations((-2, 3, -5, 4, 1)))`	11	11	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Write a program to eliminate the common elements in the given 2 arrays and print only the non-repeating elements and the total number of such non-repeating elements.

Input Format:

The first line contains space-separated values, denoting the size of the two arrays in integer format respectively.

The next two lines contain the space-separated integer arrays to be compared.

Sample Input:

5 4

1 2 8 6 5

2 6 8 10

Sample Output:

1 5 10

3

Sample Input:

5 5

1 2 3 4 5

1 2 3 4 5

Sample Output:

NO SUCH ELEMENTS

For example:

Input	Result
5 4 1 2 8 6 5 2 6 8 10	1 5 10 3
5 5 1 2 3 4 5 1 2 3 4 5	NO SUCH ELEMENTS

Answer: (penalty regime: 0 %)

```
import sys

def main():
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    # Parse sizes n and m
    n = int(input_data[0])
    m = int(input_data[1])

    # Extract elements for both arrays
    arr1 = [int(x) for x in input_data[2 : 2 + n]]
    arr2 = [int(x) for x in input_data[2 + n : 2 + n + m]]

    set1 = set(arr1)
    set2 = set(arr2)

```

```
    # Identify common elements present in both arrays
    common = set1 & set2

    # Collect non-common elements while preserving original order
    non_repeating = []

    for x in arr1:
        if x not in common and x not in non_repeating:
            non_repeating.append(x)

    for x in arr2:
        if x not in common and x not in non_repeating:
            non_repeating.append(x)

    if non_repeating:
        print(*non_repeating)
        print(len(non_repeating))
    else:
        print("NO SUCH ELEMENTS")

if __name__ == "__main__":
    main()
```

	Input	Expected	Got	
✓	5 4 1 2 8 6 5 2 6 8 10	1 5 10 3	1 5 10 3	✓
✓	3 3 10 10 10 10 11 12	11 12 2	11 12 2	✓
✓	5 5 1 2 3 4 5 1 2 3 4 5	NO SUCH ELEMENTS	NO SUCH ELEMENTS	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Given a tuple and a positive integer k, the task is to find the count of distinct pairs in the tuple whose sum is equal to **K**.

Examples:

> **Input:** t = (5, 6, 5, 7, 7, 8), K = 13
> **Output:** 2
> **Explanation:**
> Pairs with sum K(= 13) are {(5, 8), (6, 7), (6, 7)}.
> Therefore, distinct pairs with sum K(= 13) are { (5, 8), (6, 7) }.
> Therefore, the required output is 2.

For example:

Input	Result
1,2,1,2,5 3	1
1,2 0	0

Answer: (penalty regime: 0 %)

```
import sys

def main():
    lines = sys.stdin.read().splitlines()
    if not lines:
        return

    # Read tuple elements
    raw_nums = lines[0].strip()
    if raw_nums:
        nums = [int(x.strip()) for x in raw_nums.split(',')]
    else:
        nums = []

    # Read K
    k = int(lines[1].strip())

    seen = set()
    distinct_pairs = set()

    for num in nums:
        target = k - num
        if target in seen:
            # Store pairs in sorted order (smaller, larger) to guarantee uniqueness
            pair = (min(num, target), max(num, target))
            distinct_pairs.add(pair)
        seen.add(num)

    print(len(distinct_pairs))

if __name__ == "__main__":
    main()
```

	Input	Expected	Got	
✓	5,6,5,7,7,8 13	2	2	✓
✓	1,2,1,2,5 3	1	1	✓
✓	1,2 0	0	0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

The **DNA sequence** is composed of a series of nucleotides abbreviated as `'A'`, `'C'`, `'G'`, and `'T'`.

- For example, `"ACGAATTCCG"` is a **DNA sequence**.

When studying **DNA**, it is useful to identify repeated sequences within the DNA.

Given a string `s` that represents a **DNA sequence**, return all the **`10`-letter-long** sequences (substrings) that occur more than once in a DNA molecule. You may return the answer in **any order**.

Example 1:

```
Input: s = "AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT"
Output: ["AAAAACCCCC","CCCCCAAAAA"]
```

Example 2:

```
Input: s = "AAAAAAAAAAAAA"
Output: ["AAAAAAAAAA"]
```

For example:

Input	Result
AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT	AAAAACCCCC CCCCCAAAAA

Answer: (penalty regime: 0 %)

```
import sys

def find_repeated_dna_sequences(s: str):
    seen = set()
    repeated = set()

    # Extract every 10-letter substring
    for i in range(len(s) - 9):
        sub = s[i:i+10]
        if sub in seen:
            repeated.add(sub)
        else:
            seen.add(sub)

    return list(repeated)

if __name__ == "__main__":
    s = sys.stdin.read().strip()
    if s:
        results = find_repeated_dna_sequences(s)
        for seq in results:
            print(seq)
```

	Input	Expected	Got	
✓	AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT	AAAAACCCCC CCCCCAAAAA	AAAAACCCCC CCCCCAAAAA	✓

	Input	Expected	Got	
✓	AAAAAAAAAAAA	AAAAAAAAAA	AAAAAAAAAA	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` containing `n + 1` integers where each integer is in the range `[1, n]` inclusive.There is only **one repeated number** in `nums`, return *this repeated number*. Solve the problem using set.

Example 1:

```
Input: nums = [1,3,4,2,2]
```

```
Output: 2
```

Example 2:

```
Input: nums = [3,1,3,4,2]
```

```
Output: 3
```

For example:

Input	Result
1 3 4 4 2	4

Answer: (penalty regime: 0 %)

```
import sys

def find_duplicate():
    # Read integers from standard input
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    nums = [int(x) for x in input_data]

    seen = set()
    for num in nums:
        if num in seen:
            print(num)
            return
        seen.add(num)

if __name__ == "__main__":
    find_duplicate()
```

	Input	Expected	Got	
✓	1 3 4 4 2	4	4	✓
✓	1 2 2 3 4 5 6 7	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Check if a set is a subset of another set.

Example:

Sample Input1:

mango apple

mango orange

mango

output1:

yes

set3 is subset of set1 and set2

```
input2:
```

```
mango orange
```

```
banana orange
```

```
grapes
```

```
output2:
```

no

For example:

Test	Input	Result
1	mango apple mango orange mango	yes set3 is subset of set1 and set2
2	mango orange banana orange grapes	No

Answer: (penalty regime: 0 %)

```
import sys

def main():
    lines = sys.stdin.read().splitlines()
    if not lines:
        return

    # Parse the three sets from input lines
    set1 = set(lines[0].split())
    set2 = set(lines[1].split())
    set3 = set(lines[2].split())

    # Check if set3 is a subset of set1 and set2
    is_subset1 = set3.issubset(set1) # or set3 <= set1
    is_subset2 = set3.issubset(set2) # or set3 <= set2

    if is_subset1 and is_subset2:
```

```
        print("yes")
        print("set3 is subset of set1 and set2")
    elif is_subset1:
        print("yes")
        print("set3 is subset of set1")
    elif is_subset2:
        print("yes")
        print("set3 is subset of set2")
    else:
        print("No")

if __name__ == "__main__":
    main()
```

	Test	Input	Expected	Got	
✓	1	mango apple mango orange mango	yes set3 is subset of set1 and set2	yes set3 is subset of set1 and set2	✓
✓	2	mango orange banana orange grapes	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Coders here is a simple task for you, Given string str. Your task is to check whether it is a binary string or not by using python set.

Examples:

Input: str = "01010101010"

Output: Yes

Input: str = "REC101"

Output: No

For example:

Input	Result
01010101010	Yes
010101 10101	No

Answer: (penalty regime: 0 %)

```
import sys

def main():
    s = sys.stdin.read().rstrip('\r\n')

    # Extract unique characters present in the string
    char_set = set(s)

    # Check if all characters are a subset of {'0', '1'}
    if char_set <= {'0', '1'}:
        print("Yes")
    else:
        print("No")

if __name__ == "__main__":
    main()
```

	Input	Expected	Got	
✓	01010101010	Yes	Yes	✓
✓	REC123	No	No	✓
✓	010101 10101	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.